PCMR EM SCHOOL

SUMMATIVE ASSESSMENT - 3

Name :	
Class :	8^{th}
Section :	ALL
Subject :	MATHEMATICS
Marks : 100	Time : 3 Hours

I. Answer any five of the following **(5 x 8 = 40M)**

1. **Carry out the following divisions:**
   - $i)\ 28x^4 \div 56x$
   - $ii)\ -36y^3 \div 9y^2$
   - $iii)\ 66pq^2r^3 \div 11qr^2$
   - $iv)\ 34x^3y^3z^3 \div 51xy^2z^3$
2. **Factorize:**
   - $i)\ 4p^2 - qp^2$
   - $ii)\ 63a^2 - 112b^2$
   - $iii)\ 49x^2 - 36$
   - $iv)\ 16x^5 - 144x^3$
3. **Solve:**
   - i) A machine in a soft drink factory fills 840 bottles in six hours. How many bottles will it fill in five hours?
   - ii) In a model of a ship, the mast is 9 cm high, while the mast of the actual ship is 12 m high. If the length of the ship is 28 m, how long is the model ship?
4. **Find the value of:**
   - $i)\ (3^0 + 4^1) \times 2^2$
   - $ii)\ (2^{-1} \times 4^{-1}) \div 2^{-2}$
   - $iii)\ \left(\frac{1}{2}\right)^{-2} + \left(\frac{1}{3}\right)^{-2} + \left(\frac{1}{4}\right)^{-2}$
   - $iv)\ (3^{-1} + 4^{-1} + 5^{-1})^{\circ}$
5. i) The diagonals of a rhombus are 7.5 cm and 12 cm. Find its area.

   ii) The area of a trapezium is 34 cm^2 and the length of one of the parallel sides is 10 cm and its height is 4 cm. Find the length of the other parallel side.

6. **Draw the graph for the following table of values, with suitable scales on the axes.**

Cost of apples:

Number of apples	1	2	3	4	5
Cost (in Rs.)	5	10	15	20	25

II. Answer the following questions (8x4 = 32M)

7. **Solve the following linear equation:**

$$\frac{x+5}{3} = \frac{x+3}{5}$$

8. The measures of two adjacent angles of a parallelogram are in the ratio 3:2. Find the measure of each of the angles of the parallelogram.
9. Find the square root of the number by prime factorization method: **4096**.
10. Subtract 4a - 7ab + 3b + 12 from 12a - 9ab + 5b - 3.
11. Find the product of the following pairs of monomials:
    - a) 4, 7p
    - b) -4p, 7p
    - c) -4p, 7pq
    - d) $4p^3$, -3p
12. When a die is thrown, list the outcomes of an event of getting:
    - i) A prime number
    - ii) Not a prime number
    - iii) A number greater than 5
    - iv) A number not greater than 5
13. Write a Pythagorean triplet whose one member is:
    - a) 6
    - b) 14
14. Find the cube root of the following number by prime factorization method: **10648**.

III. Answer the following questions (8 x 2 = 16M)

15. What is a regular polygon? State the name of a regular polygon of:

    a) 3 sides b) 4 sides

    c) 6 sides

16. Find the square of the numbers:

- i) 32
- ii) 50

17. Find x value in $x = \frac{4}{5}\,(x\ +\ 10)$

18. Solve the following equation: 5t − 3 = 3t − 5.

19. Convert the following ratios to percentages:

a) 3:4 b) 2:3

20. Add the following:

ab - bc, bc - ca, ca - ab

21. Factorize the following expressions:

i) 7x + 42 ii) 6p – 12q

22. Evaluate:

i) 3^{-2} ii) $(4)^{-2}$

IV. Answer the following (12 x 1 = 12M)

23. General form of linear equation in one variable _______.
24. Number of measurements required to construct a quadrilateral is _______.
25. Range = _______.
26. Square root of 576 ($\sqrt{576}$) is _______.
27. $a^m \times\ a^n$ = _______.
28. Write the formula of volume of cuboid _______.
29. Find x value in 4x = 3x + 5.
30. Rational numbers are denoted by _______.
31. An algebraic expression containing three terms is called _______.
32. $\sqrt[3]{125}$ (Cube root of 125) is _______.
33. Find the value of 2^{-3} = _______.
34. x_1 and x_2 are the values of x corresponding to the values y_1 and y_2 of y, then direct proportion _______.